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Carbon dioxide liquefaction and transport optimization for marine carbon sequestration: A comprehensive review

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ABSTRACT

As global demand to reduce carbon emissions intensity, the liquefaction and transportation stages of ocean-based CO_2 sequestration have become major bottlenecks in system energy consumption and overall economic feasibility. This review develops an integrated analytical framework to examine optimization strategies for CO_2 liquefaction and transportation. It synthesizes progress in multi-stage liquefaction cycle structures, alternative working fluid, and thermodynamic integration schemes, and compares the performance, energy consumption, safety, and application scenarios of liquid (LCO_2), gaseous (GCO_2), and solid (SCO_2) transportation modes. The review also evaluates advances in boil-off gas management, including onboard reliquefaction, ammonia refrigeration, and hybrid refrigeration cycles, and highlights the synergistic use of LNG cold energy, industrial waste heat, and solar energy. Furthermore, emerging operational approaches such as multi-objective optimization and AI-enabled predictive control are assessed for their potential to enhance adaptability and dynamic system stability. Across the literature, the review identifies consistent trends toward multi-energy gradient utilization, modular integration, and intelligent scheduling, while also highlighting gaps in full-chain techno-economic assessment and large-scale demonstration data. Overall, the findings provide a structured understanding of current technologies and outline key research directions to support the large-scale deployment of CO_2 ocean sequestration.

1. Introduction

As a primary greenhouse gas, CO_2 continues to accumulate in the atmosphere and intensify global warming. The resulting impacts on ecosystems, human societies, and the global economy are substantial. The 2025 Energy Review Report indicates that energy-related CO_2 emissions increased by 0.8% [1–3]. Therefore, reducing atmospheric CO_2 concentrations is a critical objective [4–6]. Key mitigation pathways include (i) Accelerating renewable deployment and improving energy efficiency through technological and operational optimization to curb emissions at the source, and (ii) Advancing carbon capture and storage (CCS) technologies to capture CO_2 from industrial emissions and store it permanently in deep geological formations onshore or offshore, thereby preventing atmospheric release [7–9].

In 1977, Marchetti [10] first proposed the concept of CO_2 ocean sequestration, which provide a new avenue of research. Because the ocean cover over 71% of Earth's surface and the deep sea provides sufficient space to accommodate vast quantities of CO_2. Oceanic storage offers greater capacity and stronger sequestration capabilities than Land-based sequestration [11,12]. Moreover, the geological structure of the seabed can effectively prevent CO_2 leakage [13–15]. Consequently, ocean-based sequestration has enormous potential for addressing climate change.

In addition to long-term geological storage, CO_2 sequestration is increasingly being explored for its use in enhanced oil recovery (EOR) and enhanced gas recovery (EGR). The captured carbon dioxide is usually compressed into a liquid state and transported efficiently via pipelines or ships to the offshore injection sites [13,16]. After being injected into depleted offshore oil and gas fields, it can not only increase the recovery rate but also achieve permanent storage [17,18]. This highlights the dual benefits of CO_2 sequestration: both mitigating climate change through storage and improving energy production through EOR/EGR processes. Recent advances demonstrate the synergistic use of liquid CO_2 within the natural gas value chain. Kim et al. [19] proposed a CSEGR framework using liquid CO_2 mixed refrigerant (LCO_2-MR) cycles, which recover cold energy from LNG regasification to

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drive natural gas liquefaction while injecting CO_2 for enhanced gas recovery. Expanding on this, Kim et al. [20] integrated cryogenic CCUS with power-to-gas technologies for net-zero CO_2 economies. While both studies focus on cold energy synergy, Kim D et al. [19] examine conventional LNG-CCS coupling, whereas Kim Y et al. [20] extend this concept to renewable-driven systems. Both highlight the need for multi-objective optimization, pointing out the absence of a systematic review in oceanic CCS applications. In 1996, the Sleipner oil field in the Norwegian North Sea launched the world's first industrial-scale CO_2 geological storage project. This marked the transition of CCS technology from proof-of-concept to practical application. A series of long-term monitoring campaigns has verified stable storage, confirming the technology's practicality [21,22].

However, achieving large-scale ocean sequestration requires more than advances in capture and injection sections. In terms of capacity, a single CO_2 plume in deep saline aquifers can store several million tons of CO_2 [23]. Under typical geological sequestration conditions, CO_2 is injected as a supercritical or liquid-like fluid. This state has a much higher density than gaseous CO_2, allowing a greater mass to be transported and stored per unit volume of pore space [24,25]. As shown in Fig. 1, the liquefaction and transportation of CO_2 are the critical link between source capture and final storage. For long-distance, multi-national, and transoceanic situations, and for coastal or island regions lacking suitable geological conditions [26–28], the feasibility and economics of the system are determined by CO_2 liquefaction and transportation. Previous studies have approached this topic from various perspectives, including optimization of liquefaction energy consumption, shipping technology and safety, as well as storage stability and environmental risks [29–31].

Feng and Tatiana [32] evaluated the energy and economic aspects of CO_2 liquefaction and the specific electricity demand was estimated to be 60-120 kWh/t CO_2, exceeding that of capture and compression. Bennæs et al. [33] noted that at low-to-medium capture scales, shipping has higher economic feasibility than pipeline transportation due to its flexibility. However, as scale increases, the proportion of transportation costs in total cost decreases from about 22% to 10%. It indicates that the transportation should be balanced with project scales. Vakili et al. [34] emphasize that capital expenditures for marine transportation, vessel design, and route planning are decisive for the economic viability of CCS projects, as they directly determine transport costs and efficiency. Cha et al. [35] proposed a long-distance, large-scale CCS model based on liquefied CO_2 shipping from a holistic value chain perspective. The study systematically the determinants of system success. It examined liquefaction conditions, vessel selection, business models, and cross-border regulation. It also emphasized persistent constraints, including high energy consumption, limited economic viability, and the absence of international regulations. Therefore, optimizing the liquefaction and transportation processes not only helps reduce costs but also significantly enhances the economic viability of the system, particularly in transnational and deep-sea storage applications. However, it is observed that existing research mainly focuses on single dimensions, such as the thermodynamic efficiency of liquefaction energy consumption, economies of scale in transportation, or the impact of vessel design on costs, there is a lack of systematic review of the synergistic interactions and systemic coupling among these three factors. This represents a deficiency in current research and the significance of undertaking this study.

In conclusion, existing research on CO_2 liquefaction and transportation has made positive advances in energy efficiency improvements, transportation mode selection, and energy utilization. However, it still lacks a comprehensive and integrated approach. In the context of ocean carbon sequestration, the issue of multi-stage coordinated optimization has not been sufficiently explored.

2. Thermodynamic fundamentals and engineering pathways for CO_2 liquefaction

In the research on the synergistic optimization of CO_2 liquefaction and transportation systems, an in-depth analysis of the liquefaction process itself forms the basis for optimization studies. This chapter focuses on the thermodynamic principles and engineering pathways for CO_2 liquefaction, emphasizing its thermodynamic properties. It systematically illustrates the physical properties and phase characteristics of CO_2, providing a foundation for understanding and comparing the design principles, energy efficiency, and optimization potential of various liquefaction cycle systems.

2.1. *Physical properties and phase characteristics of CO_2*

CO_2, as a common industrial by-product gas, has thermophysical properties that play a crucial role in the design of liquefaction systems. The triple point of CO_2 occurs at approximately −56.6 °C and 5.18 bar, while its critical point is at 31.1 °C and 73.8 bar [36]. Fig. 2 clearly illustrates the phase transition behavior of CO_2 at different temperatures and pressures, which is essential for the precise design of the liquefaction process. Li et al. [37] conducted a systematic evaluation of the

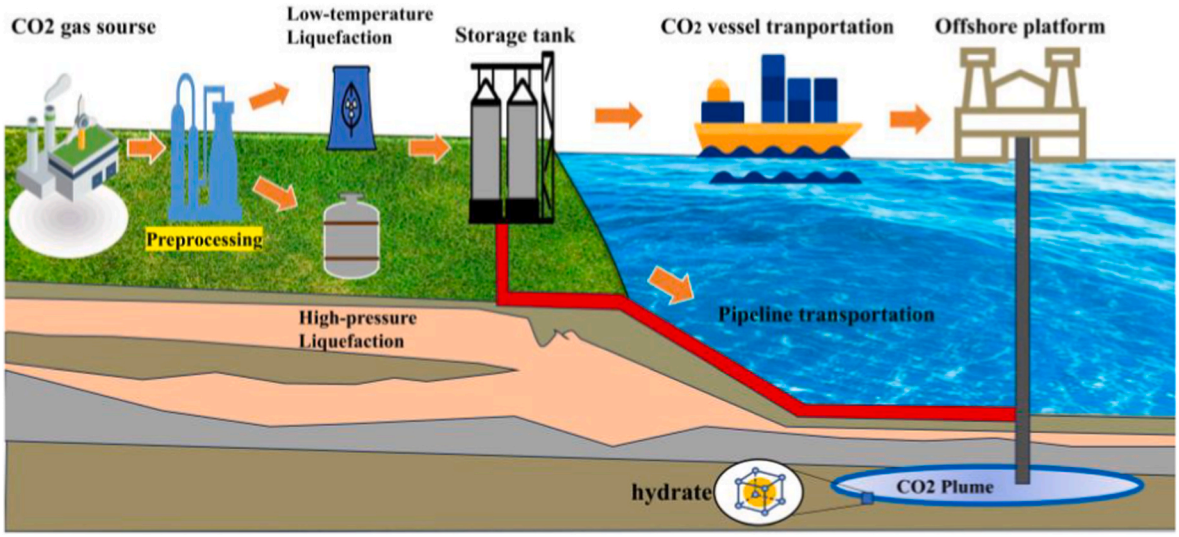


Fig. 1. Simplified schematic diagram of the entire process: Carbon dioxide liquefaction, transportation, and ocean sequestration.

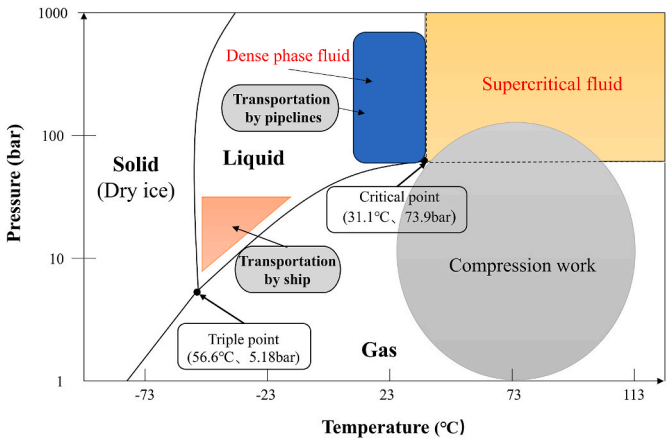


Fig. 2. Carbon dioxide phase diagram and physical properties with phase characteristics.

PVTxy properties of CO_2 and its mixtures, emphasizing the necessity of accurately verifying property models within the CCS operational windows to reduce errors in the design process. The study revealed that prediction errors for physical property models increase significantly by about 10-20% in the critical and inversion regions. Furthermore, the gas-liquid balance behavior of mixtures deviates from that of pure CO_2. This indicates that physical properties in these phase regions exhibit high sensitivity, which significantly impact the design and stability of liquefaction processes. Stian et al. [38] conducted experimental simulations revealing that when operating conditions approach the triple-phase pressure, CO_2 is highly prone to gas-solid phase transitions and the formation of dry ice. This poses a potential threat to the operational safety of liquefaction equipment. Thus, when designing the liquefaction path, it is essential to consider how to effectively control temperature and pressure changes to avoid such adverse phase transitions.

In addition to the phase change characteristics, the thermal physical properties of CO_2 also exhibit significant expansion effects. Among these, the Joule-Thomson (J-T) effect plays a central role in the design of the liquefaction process. The J-T effect describes the temperature change in a gas during adiabatic expansion, and it directly depends on the gas's phase state, temperature, and pressure. Gernert and Span [39] proposed the EOS-CG model based on the Helmholtz free energy. This model more accurately describes the J-T behavior of CO_2 mixtures and provides a reliable framework for thermodynamic calculations in liquefaction processes. Gao et al. [40] experimentally confirmed that under normal temperature conditions, the CO_2-N_2 mixture exhibits a positive J-T coefficient, enabling effective pre-cooling in open throttling liquefaction processes. their findings verify the feasibility of the model's predictions and further highlight the potential of this effect in reducing energy consumption. On this basis, researchers have attempted to integrate the J-T effect into the systematic optimization of the liquefaction process. Ariadji et al. [41] refined the J-T equation, revealing its sensitivity to CO_2 liquefaction efficiency under supercritical conditions and emphasizing the importance of temperature and pressure control. Fig. 3 shows that an increase in temperature leads to a decrease in the expansion coefficient, which in turn reduces liquefaction efficiency. Additionally, CO_2 concentration significantly affects the expansion coefficient. The modified equation shows high agreement with the EOS·PR model at a 35% CO_2 concentration, demonstrating its high accuracy and adaptability. Costa [42] combined the J-T effect with other liquefaction processes to propose a cooperative utilization scheme. This approach not only enhanced liquefaction efficiency but also offered new insights for optimizing the economic viability of the CCS value chain. In short, existing research has from theoretical modeling and experimental verification to equation refinement and system integration, demonstrating the significant role of the J-T effect in optimizing CO_2 liquefaction. However, accurately predicting and applying the J-T effect under varying impurity contents and operating conditions remains a challenge for future studies.

After understanding the phase characteristics and expansion effects of CO_2, it is essential to further investigate its unique thermal properties and flow behavior under critical and supercritical conditions, as these are crucial for ensuring the stability and efficiency of the liquefaction process. Under these conditions, the thermal physical properties of CO_2 exhibit nonlinear characteristics. Multiple experimental studies have shown that its specific heat capacity at constant pressure peaks near the critical temperature, leading to sudden changes in heat exchange performance and energy demand. Min et al. [43] systematically indicated

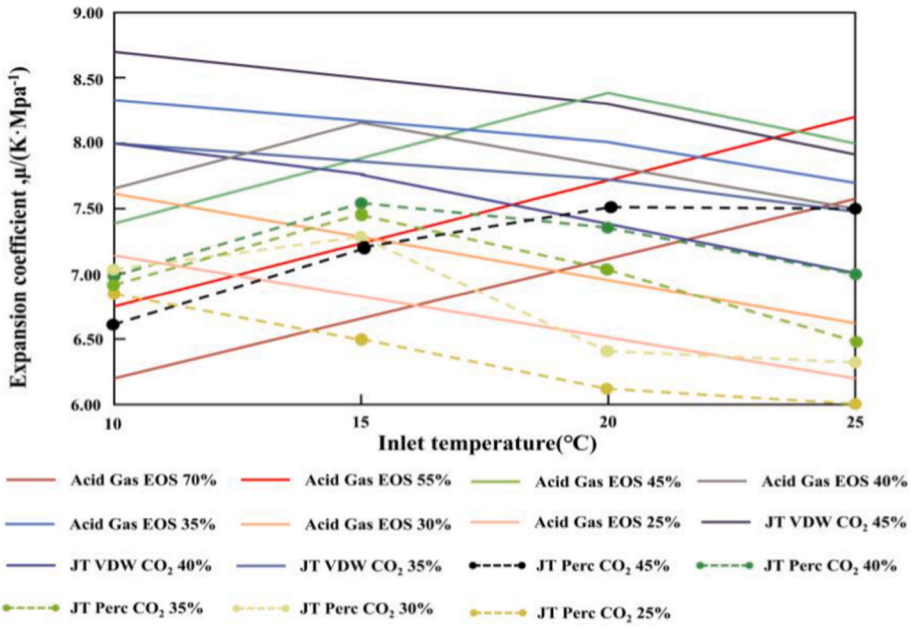


Fig. 3. Relationship graph between the coefficient of expansion and gas temperature under different carbon dioxide concentrations using different state equations [41].

that flow instabilities exist during the transition between subcritical and supercritical regions of CO_2 in transportation and injection processes. They proposed design conditions to maintain safety and stability in response to this issue. The horizontal circular tube experiments conducted by Wang et al. [44] further validated these findings. However, under suitable flow and heat transfer conditions, partial heat transfer enhancement can still be achieved, suggesting that this region is not merely a risk zone. Meanwhile, Lyu et al. [45] found in the mini-channel experiments that the heat transfer coefficient increased significantly when the temperature of the working medium approached the pseudo-critical point. Although this phenomenon increased the cooling load, it provided new possibilities for heat transfer optimization and energy recovery. Wang et al. [46] demonstrated through transient modeling based on the finite volume method that when the wall heat flows exceed a critical threshold, the transition of CO_2 from subcritical to supercritical conditions triggers significant fluctuations in density and mass flow rate. This, in turn, induces self-sustained oscillatory flow and heat transfer instabilities.

Research on the physical prosperities of CO_2 has evolved progressively equations of state and phase boundaries, to analyzing phase transition risks and process dynamics, and finally to exploring nonlinear effects in near-critical regions and system optimization. Over time, the focus has shifted from risk mitigation toward utilizing its thermodynamic properties. However, the existing research still has the following limitations: First, the quantitative models for impurity effects are still limited. Second, there is a lack of a unified predictive framework for the instability mechanisms of flow and heat transfer in the critical region. Third, the integration between physical property research and the entire process of liquefaction, transportation, and sequestration is still inadequate. Future research should balance risk control with the utilization of characteristics. It is necessary to further refine the experimental database and equations of state and promote integrated research across different stages. These efforts will support efficient and stable operation of CO_2 liquefaction and transportation systems.

2.2. Classification of liquefaction pathways

The liquefaction of CO_2 is a thermodynamic process in which temperature and pressure are controlled cooperatively. The design objective is to balance compression work and cooling capacity to ensure system safety, energy efficiency, and feasibility. In practical applications, the selection of liquefaction pathways is influenced not only by CO_2 thermophysical properties but also by factors such as transportation methods, system scale, and waste heat utilization. After understanding both physical properties and phase characteristics of CO_2, engineering practice typically categorizes liquefaction pathways into Low-temperature Low-pressure (LTLP) liquefaction and High-pressure Ambient-temperature (HPAT) liquefaction.

2.2.1. Low-temperature liquefaction

Low-temperature, low-pressure (LTLP) liquefaction is currently the most well-established technology for CO_2 liquefaction, particularly suited for long-distance transportation and offshore injection [47]. Its TRL is 8-9, ranging from system demonstration to proven commercial operation. Compared to other liquefaction processes, LTLP achieves higher CO_2 density and reduced volume, which is especially critical for marine transport and injection into underground storage. However, the optimal pressure level depends on specific operating conditions, Simon et al. [48] demonstrated that the 7 bar LTLP system, utilizing polar seawater cooling, reduces refrigeration power consumption by 18% and shortens the payback period by 1.8 years, compared to the 15 bar medium-pressure systems. Despite the latter's 11% advantage in capital expenditures (CAPEX), the 7 bar system eliminates the need for a first-stage compressor, reducing equipment costs from 42% to 35% of CAPEX [49]. Fig. 4 illustrates the economic performance of systems under different liquefaction pressures, highlighting that the 7 bar LTLP system offers significant cost savings, especially in cold climate regions. This trade-off between energy efficiency and capital cost is a common feature in LTLP system design. Lower pressure levels ($\leq$7 bar) require colder temperatures, which are economically viable only when natural cooling is available. In contrast, medium-pressure systems (10-20 bar) offer greater flexibility in geographic applicability but sacrifice energy efficiency. These findings suggest that the selection of pressure level in LTLP systems should prioritize route-specific temperature profiles rather than universal cost minimization.

Alabdulkarem et al. [50] and Øi et al. [51] found through simulation analysis that cascade refrigerant cycles can effectively reduce compression power consumption, while satisfying deep-cold requirements. Studies indicated that their efficiency gains can reach 18-25% compared to single-stage cycles. However, due to increased system complexity and stringent demands on refrigerant ratios, initial investment and maintenance costs will significantly rise. The variable-temperature bench test conducted by Jackson and Brodal [52] which quantifies the energy efficiency of LTLP liquefaction processes under varying ambient temperatures, reveals a significant impact of cooling water temperature on power consumption. The results showed that open cryogenic cycles exhibited the lowest power consumption when cooling water temperatures were below 20 °C. However, when ambient temperatures increased to 50 °C, power consumption surged by 40%, directly impacting the applicability of the LTLP processes in tropical ports. Kim et al. [53] proposed a novel low-temperature CO_2 capture and storage process integrating an LTPT CO_2 liquefaction process with a natural gas liquefaction system. By recovering cold energy from the natural gas liquefaction process, this approach significantly enhances overall energy efficiency and reduces external refrigeration loads. However, this integrated system still faces challenges in cold source adaptation and technical coordination. Collectively, these studies suggest that optimizing the LTLP process necessitates a simultaneous focus on cycle architecture, adaptation to ambient conditions, and cross-process energy integration. However, no single configuration universally optimizes all three aspects.

In conclusion, the LTLP process, as a mature CO_2 liquefaction

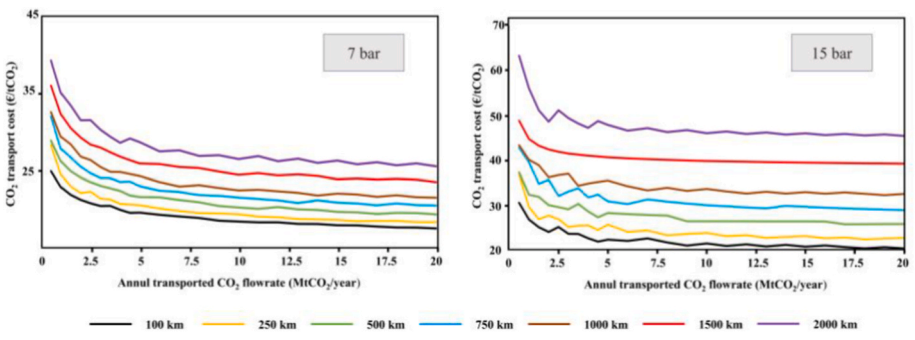


Fig. 4. Economic analysis of the system at different liquefaction pressures [48].

solution, offers high density and low vapor pressure, making it suitable for environments such as polar regions and offshore transportation. When integrated with natural gas liquefaction systems, energy efficiency can be significantly enhanced, and external refrigeration loads can be reduced through cold energy recovery. Although this approach offers advantages in specific applications, it still faces challenges. These include precision in temperature control, crystallization risks, and high initial investment costs. Future research should focus on improving refrigerant selection, enhancing heat exchange efficiency, mitigating crystallization risks, and further reducing costs through optimized system integration. Additionally, field trials under diverse climatic conditions will be crucial for driving the widespread adoption of this technology.

2.2.2. *High-pressure liquefaction*

High-Pressure Ambient-Temperature (HPAT) liquefaction provides an alternative to cryogenic cooling. The technical maturity of HPAT is lower than that of LTLP. Its TRL is 5-7 and it is still in the stage from component verification to system demonstration. It works by compressing CO_2 to 8-12 MPa at ambient temperatures, simplifying system design and improving geographic flexibility compared to LTLP processes. However, this simplification introduces distinct technical-economic trade-offs across the process chain. At the liquefaction stage, Shazawa et al. [54] demonstrated that by employing J-T throttling technology, the HPAT process can achieve 92% liquefaction rate at 15 MPa, while reducing power consumption by 14% compared to traditional system with cascade compression. This advancement makes HPAT liquefaction an effective solution for simplifying processes and lowering energy consumption.

At the transport economics level, Solomon et al. [55] developed a cost model to quantify the effects of HPAT pipeline diameter, capital investment, and interest rates on the levelized cost of Liquefied CO_2 transport (LCO_2T). As shown in Fig. 5, in the absence of booster stations, capital expenditure dominates total transport cost. for a 250 mm pipeline, operations-and-maintenance (O&M) costs also significantly impact investment decisions. These results help explain the near-term economic advantage of high-pressure liquefaction for short-distance sequestration, given the strong leverage of LCO_2T to pipeline CAPEX and financing terms. Yet material and phase stability constraints fundamentally limit this economic advantage. Experimental studies show that HPAT systems are highly sensitive to impurities. Contaminants such as N_2 and H_2O in dense-phase CO_2 significantly increase pressure drops and reduce density [56]. These phenomena include flow non-uniformity, violent phase transitions, and vibration-induced stresses. Together, they threaten pipeline structural safety and require stringent phase control [57,58]. Therefore, controlling phase transitions and impurities are not only a basis for ensuring pipeline structural safety but also crucial for preventing seal system failure and avoiding leakage incidents. Additionally, process intensification through purification represents one response to these constraints. Kim et al. [59] replaced flash tanks with distillation columns in the CO_2 compression-liquefaction process to enhance CO_2 purity. This approach significantly altered the phase behavior of the mixture, eliminating the two-phase region at pipeline transport pressures, and extending the safe operating temperature window for liquid CO_2 by 3-6 °C. However, it also led to increased system energy consumption and higher liquefaction costs. Overall, HPAT liquefaction simplifies system design. However, its limitations in compression efficiency, impurity tolerance, material constraints, and safety system design reduce its economic advantage for long-distance transportation.

In the light of these studies, the HPAT liquefaction process emerges as a promising alternative for onshore storage over medium-to-short distances and for ship transport from tropical ports. owing to its compact footprint, absence of cryogenic cooling requirements, and economic advantages in pipeline transportation. However, impurity control and the durability of sealing materials remain challenges that need to be addressed. Future research should focus on long-term demonstration experiments, particularly concerning impurity management during high-pressure liquefaction and the validation of sealing materials, to further enhance the reliability and applicability of this process.



Fig. 5. Sensitivity Analysis of Transportation Costs: The Influence of factors such as pipe diameter, investment, and interest rate [55].

2.2.3. *Comparative analysis of the two pathways*

As discussed in the previous two sections, LTLP and HPAT exhibit significant differences in their technical principles and engineering implementation, which define their distinct application cases. To systematically compare the core characteristics of these two pathways, Table 1 summarizes key operational parameters, energy efficiency, economic viability, technical challenges and so on.

Based on the comparative analysis in Table 1, LTLP and HPAT liquefaction represent two distinct technological pathways, each with its own strengths and project boundary conditions. Their selection between these pathways essentially involves balancing system complexity, energy consumption, and costs, which depends heavily on specific application scenarios and project boundary conditions. LTLP remains irreplaceable in scenarios demanding high density, such as long-distance vessel transport, meanwhile HPAT leverages its simplified system and operational cost advantages to excel in medium-to-short-distance pipeline transportation. Beyond continuous optimization within each pathway, future research must focus on developing a unified evaluation system that spans across various operating conditions. This would enable the synergistic development of both technologies by addressing their respective application boundaries, system integration, and standardization, ultimately facilitating their combined use in diverse scenarios.

Table 1
Comparative analysis of LTLP and HPAT.

	LTLP	HPAT
Operating Range	Low temperature (−20 °C)	Ambient temperature (15-35 °C)
	Low pressure (2 MPa)	High pressure (8-12 MPa)
Density features	High-density liquids; Optimized transportation efficiency	Dense-phase fluid Simplified system structure
Balancing energy efficiency	High refrigeration energy consumption;	High compression efficiency
	Cooling source issues require resolution.	Ensures stable and safe long-term operation under high-pressure impurity conditions.
System Complexity and Costs	System complexity; High initial investment;	Simplified system, Low initial investment;
	High operational and maintenance costs	Low operating costs
Challenges	High risk of crystallization and blockage	High impurity control requirements
	High temperature control requirements	High demands on sealing materials
Suitable scenarios	Long-distance vessel transportation	Medium-to-short distance pipeline transportation

3. Comparison and optimization of typical liquefaction cycle systems

Following an in-depth analysis of the thermodynamic fundamentals and engineering pathways for CO_2 liquefaction, the focus now shifts to examining the process characteristics of typical liquefaction cycle systems. This section explores how optimizing the structure and flows of liquefaction systems can improve energy efficiency and reduce costs.

3.1. Overview of a typical liquefaction cycle system process

Among CO_2 liquefaction systems, common liquefaction cycle configurations include the Linde-Hampson cycle (LH), mixed refrigerant cycle (MRC), Claud liquefaction cycle, and absorption refrigeration cycle (ARC). Each exhibits distinct advantages and limitations regarding structural complexity, thermodynamic performance, fuel efficiency, and applicable scenarios. Below is a brief introduction to four representative liquefaction cycle systems.

LH open-cycle expansion refrigeration is the classic pathway for CO_2 liquefaction. The LH liquefaction path is widely adopted due to its simple structure, low initial investment, and lack of additional refrigerant requirements. However, its single-stage coefficient of performance (COP) typically falls below 1 [60], prompting the need for performance optimization. Multi-stage throttling, pre-cooling, and high-efficiency heat exchanger design are commonly employed to improve system performance. Based on this concept, the two-stage LH has become a common variant: it first throttles and liquefies the high-pressure gas with recirculation to provide pre-cooling for the main flow, followed by a second stage of compression and throttling. This approach enhances the system's heat recovery efficiency and optimized utilization of temperature differentials. Khosravi et al. [61] present that the COP of the two-stage LH cycle is about 1.9, significantly higher than that of the single-stage cycle without pre-cooling. Fig. 6 compares the single-stage and two-stage LH liquefaction pathways, highlighting how the two-stage cycle substantially enhances liquefaction efficiency through secondary compression, pre-cooling, and multi-stage throttling. Zheng et al. [62] also demonstrated through energy efficiency and cost analysis that pre-cooled LH exhibits the best overall performance among all LH variants. However, these improvements are primarily based on steady-state process simulations, suggesting potential discrepancies between process-level COP and engineering-level system COP, necessitating further optimization research.

The ARS supplies cooling capacity through thermally driven absorption and desorption, with the common working fluids including LiBr-H_2O and NH_3-H_2O. In the CO_2 liquefaction process, the ARS is often integrated as a pre-cooling unit with transcritical or supercritical systems. By utilizing waste heat or renewable heat sources to drive refrigeration, it reduces the load on the compressor section of the system, thereby reducing its specific power consumption and enabling cascaded energy utilization [64,65]. As shown in Fig. 7, ARS achieves efficient CO_2 liquefaction through the precooling process and the thermally driven process working in tandem. In the figure, orange arrows indicate the refrigerant flow path, while yellow arrows represent the absorption and desorption processes driven by an external heat source. The

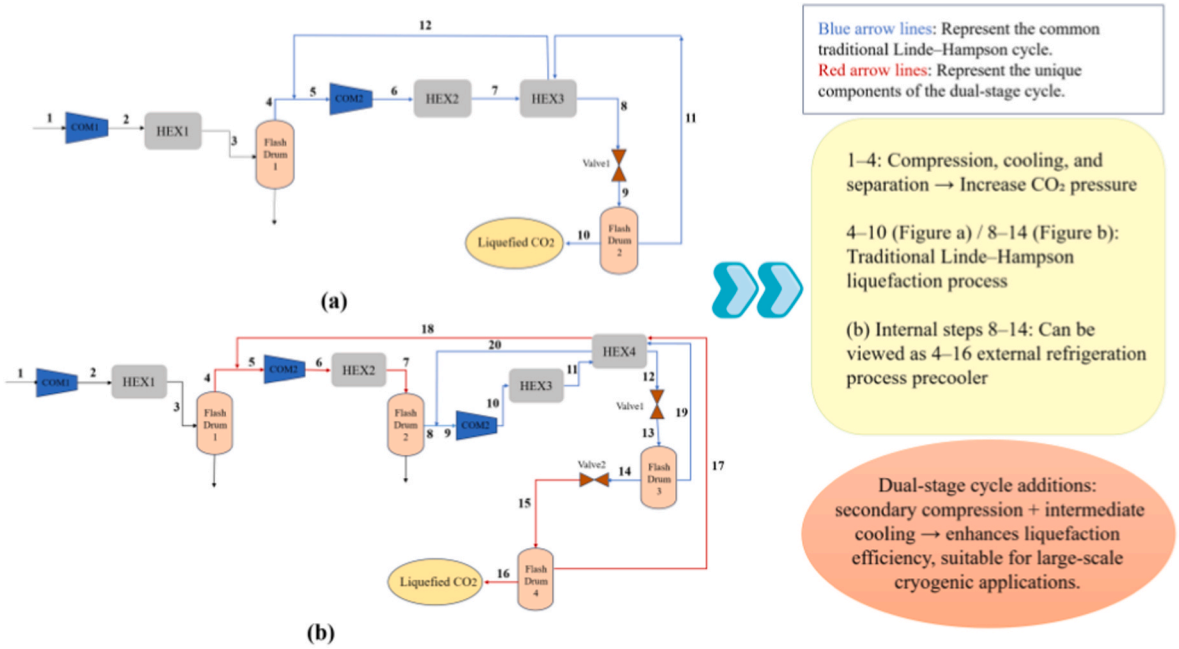


Fig. 6. Schematic comparison of single-stage and two-stage Linde–Hampson liquefaction pathways [63].

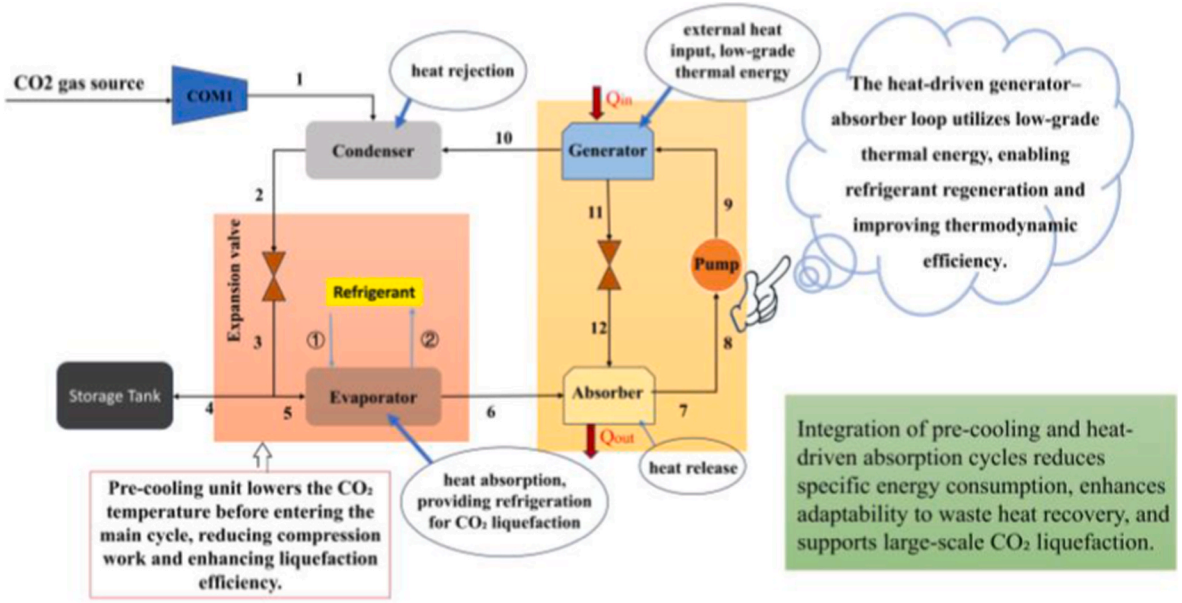


Fig. 7. Schematic diagram of ARC liquefaction pathway [70].

precooling process reduces the compressor load by lowering the gas temperature, thereby enhancing the system's overall energy efficiency. Aliyon et al. [63], Liu et al. [66], and Fei et al. [67] respectively employed steady-state simulation, constructed a 3E framework, and applied parametric optimization. Their findings show that integrating the ARS with the CO_2 liquefaction section can significantly reduce the specific liquefaction power consumption, enhance thermal efficiency, and improve overall economic performance. However, LiBr system is prone to crystallization and often operates under vacuum conditions, increasing engineering complexity. The NH_3 system requires fairing and is constrained by toxicity and material compatibility issues [68]. Moreover, ARS systems face challenges such as low COP, high maintenance costs, and large system capacity requirements [69]. However, many studies are based on ideal steady-state conditions, often neglecting fluctuations in heat sources, pressure drops, limited heat transfer area, and dynamic control considerations. Consequently, deviations may still occur between process-level benefits and engineering-level COP values.

The MRC is a closed refrigeration cycle with structural characteristics as shown in Fig. 8. The MRC system utilizes the properties of multiple refrigerants to cover a broader temperature range, making it well-suited for complex climatic conditions and large-scale liquefaction scenarios, especially in marine BOG (Boil-Off Gas) treatment. Such systems offer greater flexibility and adaptability than the LH system, providing efficient heat exchange across a broader operating range [71]. However, MRC relies on multi-stream heat exchangers, composition ratios, and multi-stage compression control, which results in a relatively complex structure and higher equipment costs. Moreover, its complexity and cost increase with the number of stages and configuration [72], and it becomes more sensitive to refrigerant formulation and fluctuations in operating parameters [73]. Only through dynamic monitoring and adjustment of composition and flow rates can optimally heat exchange matching be maintained and the specific process power kept at a low level.

The Claude cycle achieves deep gas liquefaction at lower temperatures through multi-stage compression and expansion. As shown in Fig. 9, its key processes include double compression and double cooling. Two-stage compression improves efficiency while reducing energy consumption, and the use of coolers minimizes thermal loads during compression, thereby improving overall system efficiency. Additionally, the expansion process utilizes energy recovery to expand CO_2, thereby boosting the system's energy efficiency. This approach demonstrates significantly better thermal performance and energy efficiency, making it suitable for scenarios with high energy density and stable cooling demand [75,76]. Kılıç et al. [77] and Berstad et al. [78] demonstrated through steady-state modeling and process optimization that the Claude cycle exhibits advantages in terms of specific work per unit of

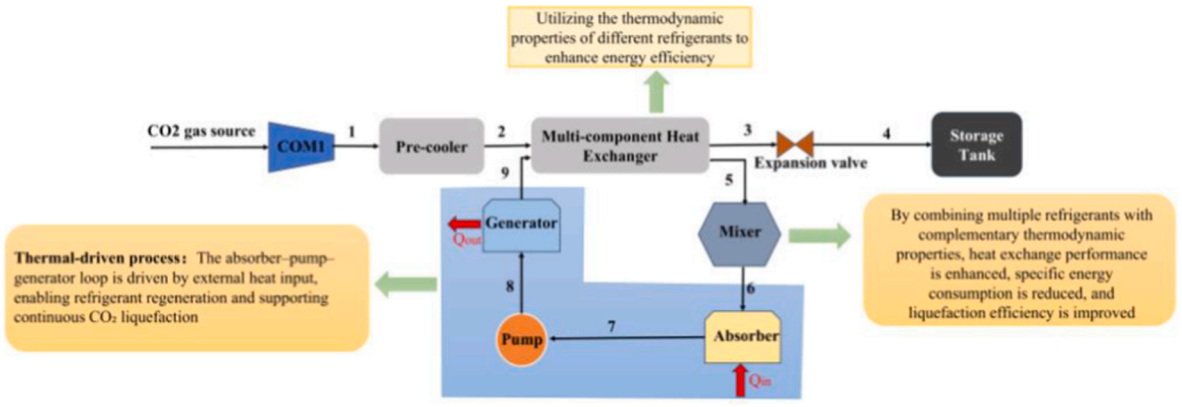


Fig. 8. Schematic diagram of MRC liquefaction pathway [74].

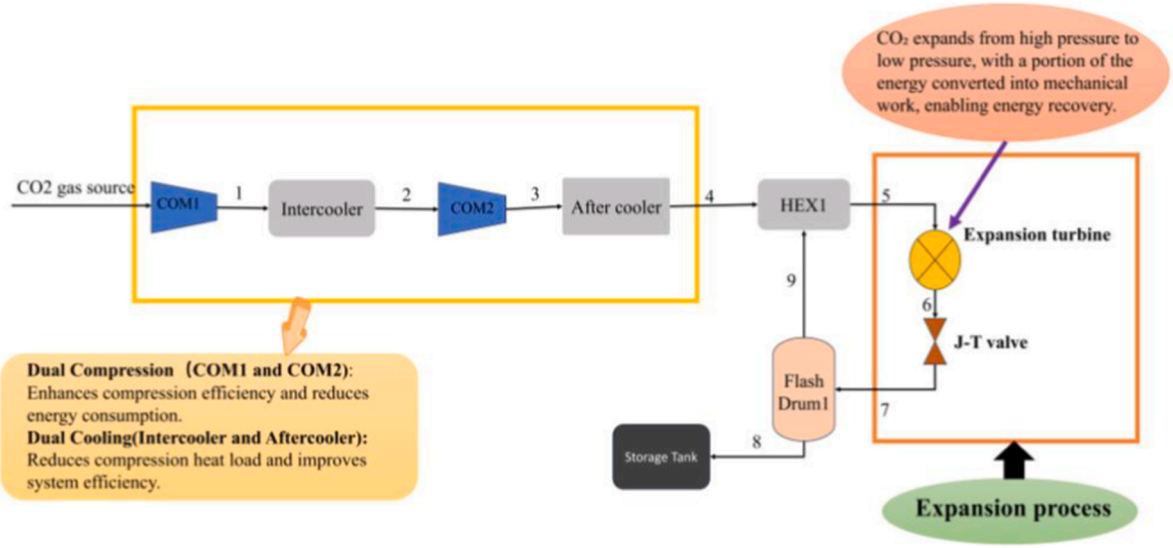


Fig. 9. Claude liquefaction process flow diagram [80].

liquefaction, system energy consumption, and cooling capacity matching. Yang et al. [79] noted that the Claude cycle is suitable for the densification of medium-to-large gas sources, particularly in high-pressure transmission or storage and transportation applications, where it offers the potential for significant energy efficiency adjustments. However, the Claude cycle features a complex structure involving multi-stage compression and expansion, demanding high-precision equipment and stringent system control, which increases engineering complexity and costs. In actual operation, limited heat transfer area, pressure drops, fluctuations, and load variations can degrade energy efficiency. Furthermore, many studies rely on ideal steady-state assumptions, often overlooking dynamic control and practical engineering factors, which may result in discrepancies between theoretical performance and real-world outcomes.

Overall, Synthesizing these cycle developments, the Technology Readiness Level (TRL) inversely correlates with efficiency potential and structural complexity. The LH cycle (TRL 8-9) is commercially mature, leveraging proven components from industrial gas applications. Research focuses on optimizing multi-stage efficiency rather than on fundamental technology development. The MRC (TRL 7-8) has been validated at the pilot scale for LNG and marine BOG applications, but CO_2-specific adaptations require further dynamic testing. The ARC (TRL 5-6) remains at the component validation stage, constrained by crystallization control, heat source variability, and low COP. The Claude cycle (TRL 5-6) is similarly under development, with challenges in turbine material compatibility and multi-stage control stability. This TRL stratification reveals a key gap: high-efficiency cycles (Claude, MRC) face engineering barriers that steady-state optimization cannot resolve, while mature cycles (LH) encounter efficiency limits that process improvements alone cannot overcome. Future research should address: (1) component-scale experimental validation of multi-stage heat exchangers and expansion turbines to advance Claude and MRC; (2) dynamic control frameworks for variable-load maritime operation; (3) hybrid cycle architectures that combine LH reliability with Claude/MRC efficiency. Overcoming these challenges will allow cycle selection based on operational context, not just efficiency metrics.

3.2. Optimization path of liquefaction system

Optimizing liquefaction systems represents a critical challenge in CCS that demands urgent solutions. Enhancing these systems not only reduces transportation costs but also improves energy efficiency. To balance energy performance and cost, numerous researchers have recently pursued a series of studies focusing on optimizing thermodynamic cycle structures during energy conversion, replacing working fluids, and integrating liquefaction systems.

3.2.1. Optimizing cycle structures

Structural optimization of the liquefaction cycle is a critical to improving efficiency and reducing transport energy consumption. Current research identifies two primary optimization strategies: multi-stage compression and heat integration, in addition to component-level customization for operational flexibility. However, these strategies have distinct applicability limits, depending on specific system requirements. Multi-stage compression consistently demonstrates a trade-off between efficiency and complexity, with diminishing returns as the number of stages increases. Engel and Kather [81] compared different-stage closed liquefaction systems and found that compared to traditional single-stage cycles, a dual-stage structure can reduce specific liquefaction energy consumption by about 39% and a triple-stage structure can further reduce it by about 13%. These results indicate that multi-stage compression with intercooling design achieves energy-saving effects and is more suitable for the high-throughput, continuous operation requirements in marine shipping scenarios.

In addition, Heat integration strategies diverge significantly in their response to environmental conditions. Hap et al. [82] and Liang et al. [83] adopted multi-parameter optimization methods for different environmental conditions, but achieved completely different thermal efficiencies. Haq et al. [82] developed a self-condensing transcritical CO_2 cycle to address condensation constraints under high-temperature conditions. The configuration uses two recuperators to recover turbine-exhaust heat and co-optimizes turbine exhaust pressure and self-condensing temperature, achieving a thermal efficiency of 39.82%. The system integrates readily with combined heat and power (CHP) schemes. Meanwhile, Liang et al. [83] proposed an adaptive CO_2 Brayton-Rankine cycle based on thermodynamic analysis and optimization, to exploit low ambient temperatures and enhance the system's thermal efficiency. Fig. 10(a) shows how turbine inlet pressure affects thermal efficiency. Color bands and isotherms highlight gains across operating conditions, and a dashed line marks the optimal pressure. Fig. 10 (b) illustrates entropy losses for each component as a function of turbine inlet pressure. As pressure increases, the system's energy efficiency and entropy efficiency exhibit different trends, demonstrating the optimization effect. The research results show that at a heat-source temperature of 278.15K, the thermal efficiency of the system reaches 50.87%, which is 2.52 percentage point higher than that of the

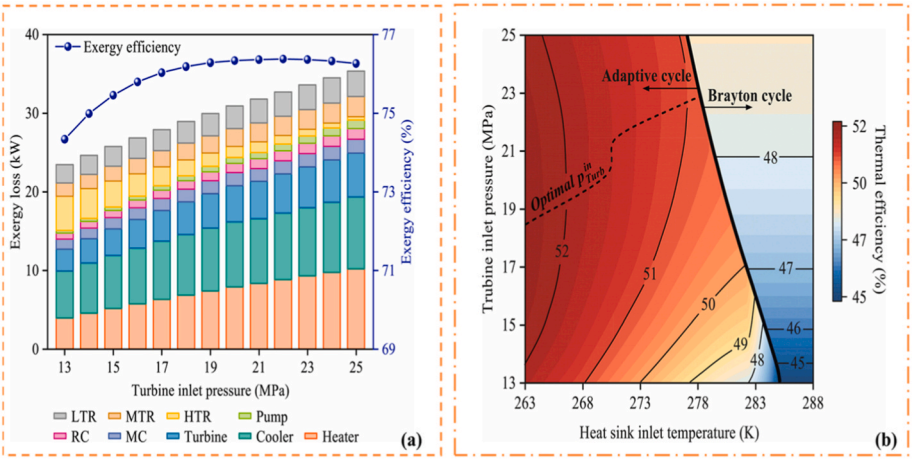


Fig. 10. (a) Sensitivity analysis and (b) parameter optimization of the adaptive CO_2 Brayton-Rankine cycle [83].

traditional supercritical CO_2 Brayton cycle. The difference of two studies indicated that optimal cycle configurations depend not only on system design but also on ambient temperature profiles and the feasibility of environmental adaptation. The concept offers a practical route to better energy use in liquefaction and scales well to larger systems.

Component-level customization aims to enhance system stability, but its application varies based on system scale. Sengupta et al. [84] and Khalili et al. [85] showed that tailoring booster compressor configurations and optimizing heat exchanger integration in ejector powered liquefaction processes can markedly improve energy efficiency and enhance stability under variable loads and ambient temperatures. The customization principles extend to hybrid systems, Sammak et al. [86] coupled a sCO_2 Brayton cycle with a semi-closed oxygen gas turbine. Their thermodynamic analysis showed that this combined system significantly improved thermal efficiency when adapting to medium-temperature heat sources. It is suitable for industrial waste gas-driven liquefaction while simultaneously meeting both liquefaction and power generation requirements. In conclusion, the optimization of the liquefaction cycle structure depends on specific conditions. The stage selection should take into account both efficiency and complexity. The optimization of pressure-temperature must be adapted to the environment, and the component design should meet the operational requirements. Future systems should integrate these strategies into a flexible design rather than focusing solely on isolated changes.

The optimization of liquefaction cycle structures has evolved from traditional throttling-compression-heat exchange methods toward multi-stage throttling, ejector booster systems, heat recovery utilization, and integration with power generation systems. These enhancements not only improve energy efficiency per unit but also strengthen system stability under variable loads and prolonged operation, providing more reliable technical support for transoceanic CO_2 transport and deep-sea storage.

3.2.2. *Optimization of working fluid substitution*

In marine applications characterized, where space is limited, stringent energy consumption must be controlled, and significant ambient temperature fluctuations are significant, traditional refrigerants have increasingly revealed issues such as low energy efficiency and substantial environmental impact. Consequently, recent research has progressively focused on low global warming potential (GWP) refrigerants, mixed refrigerants, and composite refrigerants integrated with CO_2 closed-loop systems, driving the development of liquefaction systems toward greener and lower-carbon solutions.

Current research on working fluid optimization identifies three strategies: additive-based thermophysical property modification, eco-friendly refrigerant substitution, and data-driven selection platforms, each addressing specific challenges in marine environments. Thermophysical properties of CO_2 can be adjusted through the selection of working fluids. Terracciano et al. [87] using additives to increase the critical temperature of CO_2, thereby expanding the temperature range for liquefaction operations and enhancing system stability during temperature fluctuations, which is especially crucial for oceanic CO_2 sequestration. Similarly, Yang et al. [88] used molecular dynamics simulations to examine the phase behavior and surface tension of CO_2, R32, and R134a mixtures, providing key thermophysical data for mixed refrigerants in CO_2-based systems. Their findings highlight the importance of modifying the thermophysical properties of working fluids to improve the system's efficiency and stability, which is crucial for optimizing liquefaction cycles in dynamic marine environments.

Eco-friendly refrigerant substitution exhibits divergent optimization objectives based on performance metrics and environmental targets. He et al. [89] evaluated the eco-friendly refrigerants R1234ze and R600aunder low-temperature conditions, finding that they exhibit higher COP and lower environmental impact compared to conventional refrigerants. They also demonstrated advantages in seasonal energy efficiency and annual operating costs. Fig. 11 illustrates the energy losses across different components under various thermal cycles, particularly focusing on heat exchangers and expansion valves. The analysis indicates that optimizing refrigerant selection and operating parameters can effectively reduce energy losses and enhance overall system efficiency. However, performance priorities vary significantly across studies. Qin et al. [90] through energy and exergy analysis, they found that R1234yf/R41 mixtures achieved the highest cooling capacity. In contrast, the R1234yf/R1132a mixture demonstrated the highest COP and exergy efficiency. Hou et al. [91] found that mixing CO_2 with R32 reduces the operating pressure of the liquefaction system by approximately 20%, significantly improving the overall system's performance. Their research emphasizes that the selection of refrigerants should be highly compatible with the requirements of the main system, rather than being subject to general optimization.

Beyond research on working fluid properties, some efforts focus on establishing a comprehensive working fluid data platform and optimizing the integrated operation of combined heat and power systems to further enhance the adaptability and energy efficiency of liquefaction systems in complex environments. Supporting this approach, Rogalev et al. [92] established a data platform covering the thermal properties of multiple potential refrigerants and their system performance, providing foundational support for subsequent refrigerant selection and system compatibility. Meanwhile, Wang and Hu [93] developed a data-driven method for quickly predicting the residual properties of low-GWP refrigerants. This combination of empirical databases and machine learning-based predictive tools reduces the need for extensive

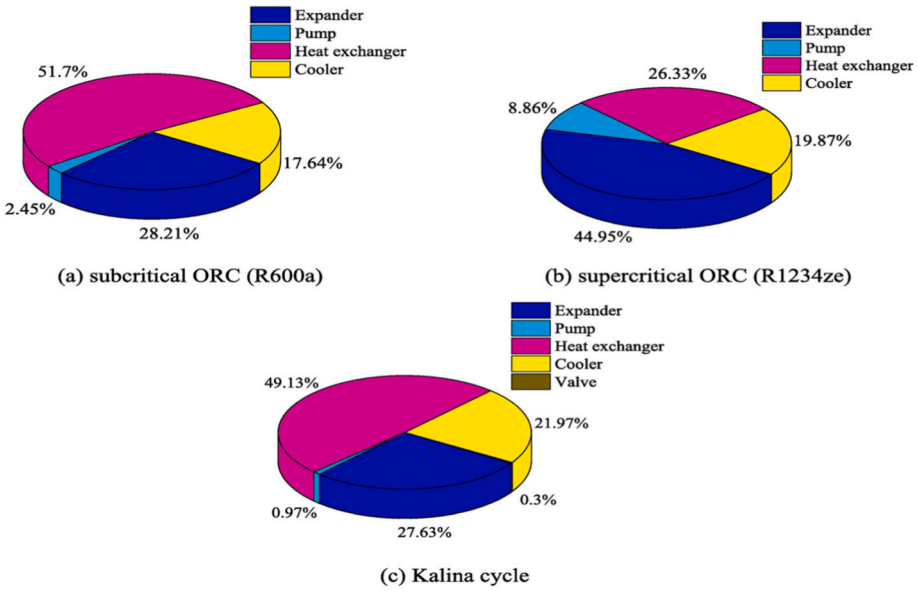


Fig. 11. Energy losses and thermal efficiency analysis of equipment under different thermal cycles [89].

experimentation. It also ensures model validation under industrial conditions, which is particularly important for marine applications, where rapid decision-making is crucial before system deployment. Chen et al. [94] introduced R161 into a combined heat, power, and cooling system. Through integrated liquefaction–hydrogen supply–cooling operations, they demonstrated the feasibility of achieving integrated energy operation with low-carbon refrigerants in complex coupled systems. Due to environmental requirements, working fluid substitution strategies have emerged as a key direction for optimizing CO_2 liquefaction systems. These substitution strategies enhance system efficiency, improve operational stability, and reduce carbon emissions, thereby providing a solid foundation for sustainable development, low-carbon marine storage liquefaction technology.

3.2.3. *System integration and process coupling*

In marine transportation and liquefaction systems, a single liquefaction process struggles to address the demands for energy efficiency, high performance, continuous refrigeration supply, and flexible scheduling at the same time. Integrated design not only helps resolve issues of space constraints and fluctuating cooling loads but also reduces shore-based energy consumption, while enhancing energy utilization efficiency on board vessels. This approach demonstrates strong adaptability and potential for widespread adoption. Recent research has focused on two converging integration pathways. One is binary coupling of CO_2 liquefaction with exergy cycles for specific performance enhancement, while the other is multi-energy subsystem integration for complex operational scenarios. However, both approaches involve trade-offs between efficiency gains and system complexity.

Binary coupling strategies leverage synergies between CO_2 liquefaction and specialized thermodynamic cycles, providing efficiency improvements but varying stability under different energy inputs. For instance, Bahram et al. [95] utilized low-grade heat to drive absorption pre-cooling, but system performance heavily depends on waste heat recovery efficiency. Dai et al. [96] coupled a transcritical CO_2 cycle with solar energy, using it to drive an absorption refrigeration unit for pre-cooling. Their annual simulations demonstrated the COP improvement ranging from 3.05% to 42.3% and a reduction in carbon emissions of nearly 40%. However, the system exhibited poor stability due to fluctuations in solar energy availability. Lee et al. [97] developed an integrated system for co-production of hydrogen and liquid CO_2 via steam methane reforming, using multi-objective optimization to improve energy and economic performance. Despite these challenges, these systems demonstrate that coupling with renewable or low-grade energy sources can enhance performance. However, their application is often limited by energy input variability.

As shown in Fig. 12, which outlines the system's energy flow and optimization framework, a genetic algorithm (GA) was applied to optimize the pre-cooling section. The results, comparing baseline, scaled-up, and integrated processes across different pressures, reveal how pressure variations influence efficiency, cost, and environmental impact. Their analysis confirms that the integrated scheme provides a valuable technical pathway by offering economic benefits alongside significant reductions in energy use and environmental footprint.

Multi-energy subsystem integration extends beyond binary coupling to address complex operational demands, though at increased system complexity and coordination costs. Mubashir et al. [98] designed a trigeneration system integrating engine exhaust recovery, sCO_2 Brayton cycle, ORC, and Absorption Refrigeration Systems (ARS) It achieved 59.2% overall efficiency and $0.063/kWh cooling electricity cost post-GA optimization. However, the system's complexity and stringent equipment coordination led to higher investment and maintenance costs. Chen et al. [99] combined sCO_2 Rankine cycles with LiBr-H_2O absorption refrigeration, demonstrating energy cascade utilization, but struggled with stability under dynamic conditions, mirroring the solar instability observed in Dai et al.'s system [96]. Additionally, Xia et al. [100] proposed a CO_2/R32 mixed refrigerant system integrated with a combined cooling and power system. This approach ensured cooling capacity and enhanced energy efficiency, which was suitable for liquefaction scenarios with limited space, such as floating platforms. However, the system's stability can be influenced by the complexities of mixture composition management and temperature transients. Yao et al. [101] proposed a combined cold energy-CO_2-steam drive pathway for LNG ship systems. By utilizing LNG cold energy for pre-cooling and waste heat to drive the evaporator, the exergy efficiency was increased to 45.1%, significantly enhancing the liquefaction depth and system loading capacity.

Synthesizing these integration strategies, CO_2 liquefaction system design faces a fundamental tension: binary coupling offers targeted efficiency gains with manageable complexity but suffers from input-specific instabilities. Multi-energy integration enhances operational flexibility and overall efficiency yet escalates coordination costs and stability risks under dynamic conditions. A shared trend across both pathways is the growing importance of intelligent optimization. GA and related algorithms are increasingly used to balance efficiency, cost, and

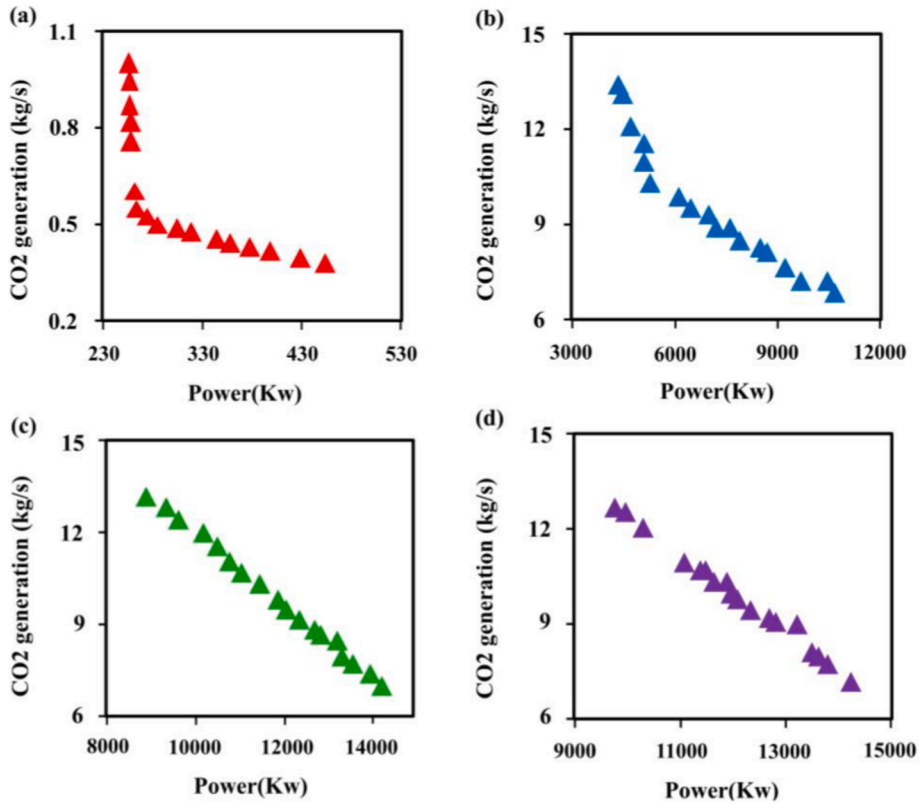


Fig. 12. Results obtained through GA optimization: (a) optimized basic process and (b) optimized amplification process, along with integrated process optimization under different pressures: (c) 80 bar and (d) 100 bar [97].

operational stability. Future work should prioritize hybrid architectures that pair stable base-load energy sources, such as waste heat and LNG cold energy, with adaptive control, rather than relying on one-size-fits-all integration schemes. With support from artificial intelligence and advanced optimization, these designs can deliver more robust and flexible solutions for low-carbon CO_2 sequestration and marine transport.

3.3. *Thermal energy integration and low-carbon energy drive strategies*

The conventional CO_2 liquefaction route, dominated by electrically driven compression and multi-stage throttling, remains constrained by high energy demand and pronounced load fluctuations. Accordingly, recent studies increasingly explore multi-source collaborative architectures that cascade external energy streams to support pre-cooling, main cooling drive, and cold-heat-power integration. Across the literature, a consistent conclusion is that the availability profile of external energy largely determines both achievable efficiency and operational stability. In general, LNG cold energy and industrial waste heat provide relatively stable, high-density inputs suitable for base-load operation, whereas renewables and distributed waste heat broaden deployment flexibility but introduce intermittency and coordination challenges. Studies integrating LNG cold energy with industrial waste heat generally report strong thermodynamic synergy because cold recovery and heat-driven power or refrigeration can be matched within a single cascade. Yousef et al. [102] proposed a dual-loop supercritical CO_2 cogeneration system that combines regasification cold energy with boiler flue-gas waste heat, achieving an overall efficiency of 62.79%. By contrast, LNG-based integration within Natural Gas Combined Cycle with Post-Combustion Capture (NGCC + PCC) frameworks is often designed around system compatibility and retrofit constraints rather than peak efficiency. Bao et al. [103] and Sultan et al. [104] show that this approach limits efficiency losses to below 5%, making it particularly suitable for shore-based liquefaction plants and LNG import terminals. Extending the same cold–heat coupling concept to mobile applications, Zhi et al. [105] developed a transcritical CO_2 parallel Rankine cycle driven by engine waste heat and LNG cold energy, and reported a power-output increase of up to about 86% under the investigated conditions. Overall, stable-source schemes are well suited for continuous operation. Their designs generally follow two directions: efficiency-oriented integration or compatibility-oriented retrofits, depending on scale and retrofit constraints.

In offshore or decentralized scenarios, renewables are often used to reduce reliance on shore power and support autonomous operation. However, intermittency remains the primary limitation on performance. Mehrpooya et al. [106] proposed a solar-driven diffusion absorption refrigeration system that eliminates the need for electrically driven compression, making it suitable for offshore or distributed liquefaction scenarios. Wang et al. [107] combined data-center waste heat with CO_2 heat pumps in urban sequestration nodes. Their results suggest that moderately stable heat sources can still contribute when upgraded by heat pumps, provided that source continuity and temperature level are sufficient. Collectively, these studies indicate that integrating renewables or distributed heat broadens deployment options. However, it requires dedicated measures to address intermittency and variable source quality.

Across both stable-source and renewable-driven pathways, auxiliary subsystems are often used to match available energy quality with liquefaction demands. Heat-pump-based schemes for mid-temperature heat sources, as explored by Dai et al. [108]and Ma et al. [109], can deliver high COP and stable operation. However, they typically remain electro-thermally hybrid rather than purely thermally driven, placing them between renewable absorption autonomy [106] and direct waste-heat integration [102]. At the high-integration end, poly generation concepts such as Rabet et al. [110] constructed an integrated multi-product system model involving TPV, SOFC, hydrogen

electrolysis, and CO_2 liquefaction unit modules to achieve combined production of electricity, green hydrogen, and CO_2 liquefaction. This provides an efficient, economical, and sustainable solution for green hydrogen production and CO_2 liquefaction, serving as a significant reference for future poly-generation system design. Importantly, thermal energy storage is increasingly discussed as a practical route to reconcile the stability–flexibility dilemma. Barberis et al. [111] proposed a sCO_2 based Thermal Integration Pumped Thermal Energy Storage (TI-PTES) system. By integrating low-grade heat sources, this system enhances energy efficiency of liquefaction system and reduces external cooling loads, offering viable solution for thermal energy recovery and low-carbon energy utilization. The T-S diagram in Fig. 13 illustrates the optimized reheat discharge layout, which is designed to maximize thermal recovery efficiency. The layout was derived from sensitivity analysis and genetic algorithm optimization, optimizing the system's pressure level with an upper limit of 250 bar. This optimization aims to enhance both thermal recovery efficiency and overall system performance. The results highlight that carefully adjusting the pressure levels within the system ensures the optimal use of available waste heat, thus improving the overall efficiency of the liquefaction process. These findings establish a theoretical foundation and demonstrate the engineering feasibility of diversifying the external energy sources that drive liquefaction processes.

Taken together, the literature indicates that multi-source CO_2 liquefaction does not yield a single optimal configuration. Instead, design choices are scenario-dependent and shaped by energy-source stability, temperature level, and integration constraints. Stable baseload schemes, especially LNG cold energy coupled with industrial waste heat, are well suited for high-capacity continuous operation. Renewable-driven systems improve autonomy but usually require storage or other measures to mitigate intermittency, while distributed waste heat is most effective in hybrid schemes when upgraded by heat pumps. Future work should move beyond single-system optimization toward cross-scenario stability control, multi-energy coordination, and integrated carbon-economic assessment that reflects real deployment constraints across shore-based plants, offshore platforms, and urban nodes.

4. Optimization of CO_2 ocean transport routes and systems

The optimization of marine CCS systems requires not only refining the liquefaction process but also integrating the transportation system. This chapter provides a comprehensive analysis of the design and optimization of CO_2 transport systems, focusing on liquid transport, vessel operation, Boil-off Gas (BOG) management and recovery technologies. It further explores the application of intelligent control and system-wide optimization scheduling. These discussions aim to elucidate the synergistic effects among these components and their complementary roles in practical implementation.

4.1. Comparison of transport methods

In oceanic CCS, the method of carbon dioxide transport plays a crucial role in the energy efficiency, safety, and economics of the entire system. Current methods primarily include low-temperature liquid (LCO_2) transport, high-pressure gas (GCO_2) transport, and solid (SCO_2) transport. These approaches differ fundamentally in their thermodynamic mechanisms, transport density, system complexity, lifecycle costs, and operational timeframes. Consequently, their suitability must be evaluated based on transport distance, scale, and the physical characteristics of the target storage site.

For high-density, long-distance scenarios, LCO_2 consistently emerges as the preferred option despite its inherent complexity. This way is Baroudi et al. [112] and Pérez-Bódalo et al. [113], have highlighted LCO_2's thermodynamic advantages for large-scale transoceanic transport. Additionally, they indicated that LCO_2 transportation can leverage operational experience from LNG to inform large-scale, long-distance transportation [34]. Recent studies have highlighted infrastructure constraints as the main barrier to practical deployment. Fan et al. [114] show that LCO_2's thermodynamic properties complicate cargo containment, operational efficiency, and safety. Vakili et al. [115] emphasize that the technical and economic feasibility of LCO_2 transport is heavily influenced by infrastructure design and scale, with high capital and operational costs posing challenges for large-scale deployment. Collectively, these studies suggest that infrastructure requirements, such as cargo systems, cost scaling, and cooling hose reliability. These are crucial for determining LCO_2 viability. These factors are more influential than thermodynamic performance alone. The choice between LTLP and HPAT pathways further modifies these infrastructure demands. Because LTLP requires high-reliability cooling hose systems, while HPAT necessitates high-pressure material compatibility and compression energy management. LTLP-LCO_2 suits moderate-climate regions with established LNG terminals and moderate offshore distances (500-1000 km), exemplified by North Sea projects leveraging shared cryogenic infrastructure. HPAT-LCO_2 is necessitated for high-ambient-temperature, long-distance scenarios lacking domestic storage. East Asian industrial corridors facing transoceanic transport to Southeast Asian sites

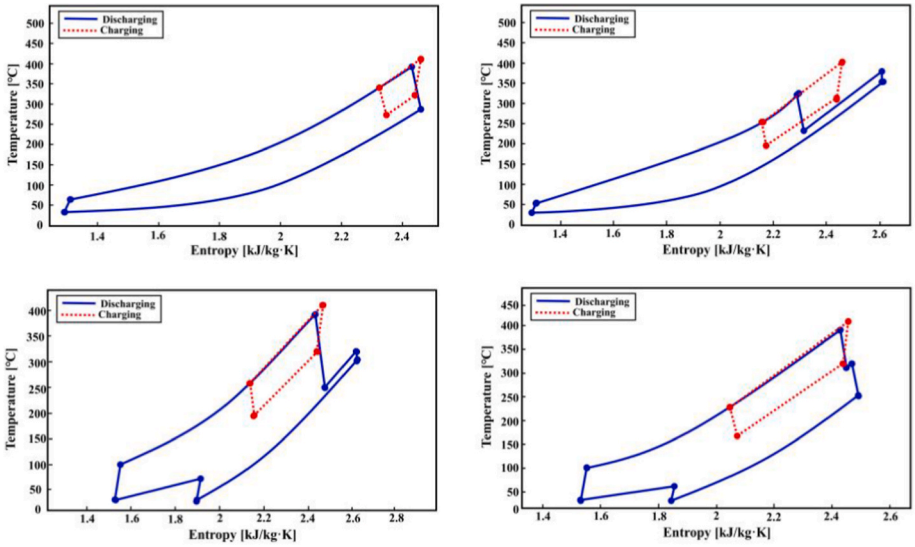


Fig. 13. T-S diagram of the optimized reheat discharge layout [111].

exemplify this: they prioritize HPAT to minimize cooling energy, accepting reduced transport density and higher unit energy demand.

GCO_2 transportation typically entails compressing CO_2 to a dense phase and via compressor stations along the pipeline, enabling stable transport at ambient temperatures. Bai and Lu [116] analyzed the impact of various impurities on pipeline transport with an integrated thermophysical and fluid flow model. Fig. 14 illustrates the effects of impurities on phase behavior, density, viscosity, and pressure loss during CO_2 pipeline transport. By categorizing and evaluating these impurities, their study provides guidance for optimizing the design and operation of CO_2 pipeline systems. Solomon [55] conducted a cost and design optimization analysis of CO_2 pipeline transportation infrastructure, concluding that GCO_2 is suitable for short-distance with stable supply, but requires the strategic placement of booster stations to minimize energy consumption. Roussanaly et al. [48] observed a significant decline in GCO2 transport efficiency with increasing distance. The energy efficiency of compression and station operation becomes the primary sources of energy consumption, thus limiting large-scale, long-distance application. Wilkes et al. [117] also emphasized the critical importance of optimizing booster station layout and compression equipment for determining the overall transport energy efficiency, noting that more precise energy efficiency assessments and system optimization are necessary during the design phase. Collectively, these studies suggest that GCO_2 viability is limited by the cumulative compression energy over distance, restricting its application to regional networks where station spacing can be optimized to meet throughput requirements. For regions with high ambient temperatures and established LNG infrastructure, such as Singapore or Middle Eastern ports, HPAT liquefaction combined with short-range GCO_2 transport offers an efficient solution. HPAT-GCO_2 is ideal for tropical ports with established LNG infrastructure and short offshore distances (<100 km), such as Singapore and Middle Eastern ports. High ambient temperatures make LTLP cooling energy-intensive, while moderate distances favor pipeline transport over vessel complexity. This configuration utilizes LNG terminal capacity for HPAT efficiency, eliminating the boil-off gas management and charter costs associated with LCO_2 transport.

SCO_2 transport primarily occurs in the form of hydrates or hydrate slurries, enabling stable storage and transportation under relatively low temperature and pressure conditions. Fernandes [118] constructed the pipeline transport model of CO_2 hydrate slurry, highlighting its higher safer profile and milder operating conditions compared to LCO_2 and GCO_2. While the process offers potential for cryogenic cold energy utilization, it also presents challenges in achieving the required hydrate formation pressure and effectively managing the exothermic heat release. Furthermore, Aminnaji et al. [119] and Fu et al. [120] experimentally demonstrated that SCO_2 transportation is constrained by slow nucleation rate, low heat transfer efficiency, and the non-Newtonian flow characteristics, which significantly limit its application in large-scale pipeline transportation. Rukh et al. [121] concluded through a review that optimizing hydrate formation rates and employing suitable promoters can effectively enhance transport efficiency, thereby addressing limitations of existing technologies, and offering a potential solution. Jiang et al. [122] investigated the phase transition characteristics and rheological properties of CO_2 double hydrate slurries, identifying significant potential for cold energy storage and transportation, particularly in regulating hydrate formation rates and stability. Bhati et al. [123] analyzed the cost-effectiveness of CO_2 hydrate formation through a techno-economic model, demonstrating that subcritical CO_2 hydrates offer significant economic advantages for carbon sequestration, particularly in cold energy storage and long-distance transportation. Overall, although current research has revealed the potential of CO_2 hydrates in transportation, especially in terms of safety, mild operating conditions and low-temperature cold energy utilization. However, it is still in the experimental stage and faces multiple technical bottlenecks in large-scale application. Future research should focus optimizing the hydrate formation process, enhancing heat transfer efficiency and fluidity, to improve the overall efficiency and feasibility of SCO_2 transportation.

The temporal dimension of CO_2 transport, including loading, maritime transit, unloading, and injection, varies across transport modes and distances, directly impacting project scheduling. LCO_2 vessel transport typically takes 2-4 weeks for transoceanic routes (>3000 km), with 1-2 days for loading/unloading and 5-10 days for injection preparation. This cycle requires buffer storage to maintain continuous capture during vessel turnaround. GCO_2 pipeline transport is faster, with 1-3 days of compression station transit over regional distances (<1000 km), but

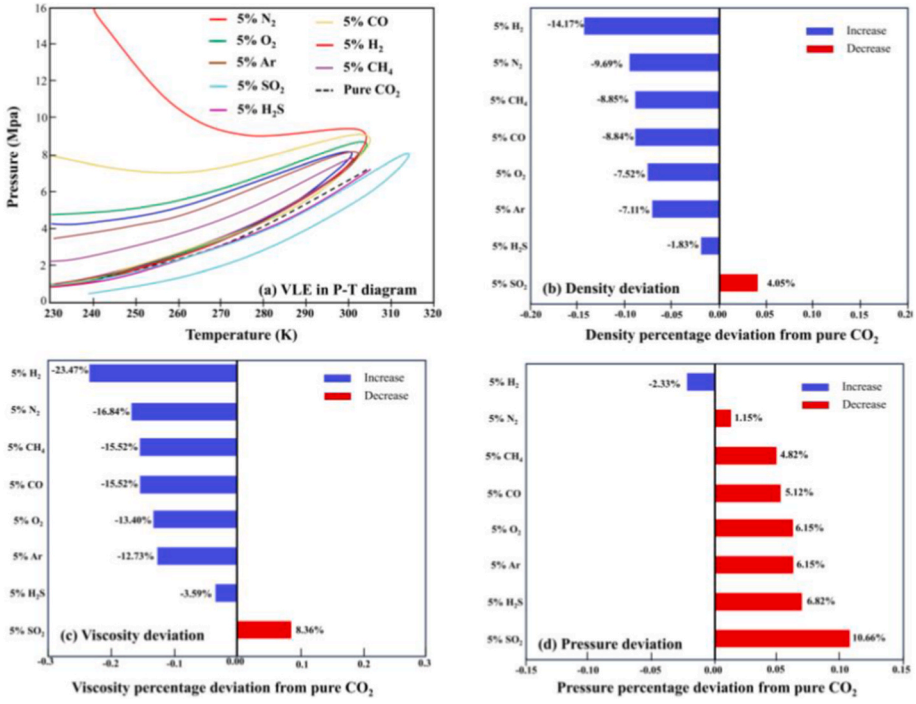


Fig. 14. Graded assessment of the impact of various impurities on GCO_2 [116].

batch injection at storage sites may still introduce 2-5 days of periodicity. SCO_2 hydrate transport, if realized, could take 1-2 weeks due to slower slurry formation but reduced injection preparation time. These durations highlight the need for scheduling models that account for vessel cycles, weather constraints, and storage site capacity. Future research should focus on dynamic scheduling and buffer storage optimization to address these operational challenges.

By comparing the advantages and disadvantages of the three CO_2 transport methods in Table 2, it is clear that: LCO_2 has become the central solution for large-scale, long-distance CCS chains due to its high transport density and energy efficiency. GCO_2, with its high technical maturity (TRL 6-7) and simplified system, is suitable for short-to-medium-distance continuous pipeline transport but requires additional compression stations. SCO_2, transported via hydrates or slurries, offers high safety and potential for cold energy utilization. However, its development is constrained by technical challenges such as slow hydrate nucleation rates and low heat transfer efficiency, with its TRL currently at 5-6. LTLP-LCO_2 is optimal for regions with established cryogenic infrastructure and moderate-to-long distances; HPAT-GCO_2 is suited for tropical ports with short offshore distances and existing LNG infrastructure; and SCO_2 remains experimental, with potential for Arctic applications. Future research should deepen in three key areas: First, establish a systematic evaluation framework for determining the applicable boundaries of transport methods; Second, promote synergistic design between transport and energy recovery, particularly focusing on cold energy utilization in liquid and solid systems and its coupling with power generation and energy storage systems; Third, accelerate breakthroughs in critical technologies such as hydrate formation and stability, optimization of slurry rheological properties, and pipeline structural design. Addressing these challenges will significantly enhance the efficiency and safety of CO_2 transport systems, reduce costs, and provide more reliable technical assurance for the practical application of marine CCS technologies.

4.2. Design and operation of marine transportation systems

With the advancement of CCS toward large-scale offshore applications, the design and operation of liquid CO_2 shipping systems have become critical factors determining the energy efficiency and economic viability of the entire chain. Existing research has explored avenues such as LNG vessel retrofitting, specialized design, energy efficiency optimization, and cold energy recovery. However, two critical gaps persist: insufficient full-chain techno-economic assessment of vessel options, and lack of large-scale demonstration data validating long-term operational reliability.

Regarding techno-economic assessment, current studies compare retrofitting versus purpose-built options using isolated cost models, without accounting for transport distance, liquefaction energy demand, or storage site characteristics. Key research directions include: (1) developing dynamic cost models that link vessel capital expenditure to regional infrastructure and route-specific operating profiles; (2) conducting pilot-scale deployments on established maritime corridors to generate operational cost data for model validation. Regarding long-term reliability, material corrosion in CO_2-rich environments and structural longevity are still unquantified at commercial scale. Specific research directions include: (1) accelerated corrosion testing for candidate materials under realistic CO_2-H_2O impurity conditions; (2) integrating structural health monitoring systems on demonstration vessels to establish predictive maintenance frameworks.

Table 3 systematically summarizes the methodologies and primary conclusions from relevant literature. Based on a comprehensive analysis of the literature listed in Table 3, current CO_2 shipping technology is transitioning from short-term retrofits of existing LNG vessels to the development of purpose-built vessels for long-term applications. Research indicates that modifying the temperature control systems and piping of LNG carrier enables their rapid conversion into LCO_2 transport platforms. However, the long-term operational sustainability and large-scale economic viability of such retrofitted systems remain subjects of ongoing evaluation. The long-term corrosion resistance of materials in CO_2-rich environments and overall structural longevity are key challenges, which collectively underscore the necessity of advancing dedicated CO_2 carrier technology.

As shown in Fig. 15, the subsea transfer system for tankers incorporates various technical elements aimed at improving transportation efficiency and stability. As transport scales expand and voyages lengthen, core design challenges have shifted from technical feasibility to ensuring safety and structural reliability during extended operations. In this context, the integration of liquefaction processes with transport systems, along with the development of high-performance materials for deep-sea environments, is central to advancing efficient and stable CO_2 maritime transport. Additionally, the management and recovery of BOG emerge as a critical factor affecting the economics and safety of long-distance transport.

4.3. BOG (boil-off gas) management and recovery

During the CO_2 transportation for ocean sequestration, the shipboard liquefaction system generates cold losses and BOG. BOG primarily forms when temperature fluctuations, vessel vibrations, and heat exchange during transport cause partial vaporization of liquid CO_2, thereby increasing the gas-phase pressure within the tanks [139]. Shipboard reliquefaction systems stabilize pressure through compression, cooling, and energy recovery. Current research focuses on three strategies for refrigeration energy supply: fuel-side cold energy recovery, optimized mixed refrigerant cycles, and power-reliquefaction integration. However, these approaches involve distinct trade-offs between efficiency gains and system complexity.Fuel-side cold energy utilization prioritizes energy recovery over external refrigeration, depending on the fuel type and level of integration. Liquid ammonia is primarily applied through two principal technical pathways: one as a circulating refrigerant, and the other as a direct cooling source. In the liquid ammonia-driven cooling route, Lu et al. [140] and Yiqun et al. [141] employed the

Table 2
Comparative analysis of three transportation methods.

Mode of transportation	Phase State	Strengths	Challenges and Limitations	Applicable Scenarios
LCO_2	Pure liquid CO_2	● High density ● High energy efficiency	● Complex system ● High energy consumption for liquefaction ● Strict temperature control requirements	large-scale and long-distance transportation
GCO_2	Dense-phase compressed CO_2	● Simple operating conditions ● Low cost	● High energy consumption of booster stations	short-distance transportation with stable supply
SCO_2	CO_2 hydrate Or Hydrate slurry	● High safety ● Mild operating conditions ● Utilization of low-temperature cold energy	● Slow nucleation rate ● Low heat transfer efficiency ● Poor fluidity caused by non-Newtonian rheological properties ● Limited large-scale application	small-to-medium-scale transportation with high safety requirements

Table 3
Summary of representative studies on CO2 shipping systems: research focuses and main conclusions.

Subject	Author	Focus of Study	Main Conclusions
1. CO_2 Terminal Techno-Economic Analysis	Lee et al. [124]	CO_2 terminal and transportation hub design model	● Foundation: System optimization of terminal design and liquefaction processes ● Advantage: Leveraging existing LNG infrastructure for synergy and cost-effectiveness
	Zahid U et al. [125]	Technical and economic assessment of CO_2 liquefaction and shipping	
	Datta A et al. [126]	Dual-purpose LNG-CO_2 vessel concept: LNG carrier retrofitted for CO_2 transport	
	Awoyomi et al. [127]	Evaluation of CO_2/SO_2 co-reduction strategies	
2. Marine Transportation System and Safety Optimization	Zhang N et al. [128]	Structural and thermodynamic analysis of shipboard CCUS technology	● Core of Safety Assurance: Synergy among tank structure, material selection, and temperature-pressure control. ● Key to Large-Scale Application: Robustness and reliability of the operational model
	Cheng H et al. [129]	Design and mechanical analysis of cryogenic hoses for LCO_2 ship-to-ship transfer	
	Choi Y et al. [130]	IMO Type C Tanks: Pressure-optimized Design for Liquid CO_2 Transport	
	Ma Y et al. [131]	Baseline design for subsea shuttle tanker systems	
3. BOG Management and Reliquefaction	Noh H et al. [132]	Technical and economic analysis of large-scale shipboard BOG reliquefaction systems	● Solution: Effective BOG management ● Key Challenge: Balancing system energy consumption, control complexity, and cost for long-haul voyages
4. Cold Energy Recovery and Process Integration	Zhang G et al. [133]	Modeling and analysis of an LNG cold energy-driven CO_2 reliquefaction system	● Core Solution: Integration of energy subsystems across the chain ● Key Benefit: Significant reduction in liquefaction energy consumption to improve economic efficiency
	Baldasso et al. [134]	Analysis of the potential for utilizing cold energy from shipborne LNG	
	Lebedevas, Malūkas [135]	Application of LNG cold energy combined with cryogenic CO_2 capture	
	Kim J, Kim D [136]	Analysis of an ammonia-fueled cold-energy-coupled ORC system	
5. CO_2 Purity and Energy Efficiency Assessment	Jung et al. [137]	Comparison of CO_2 compression and liquefaction processes	● Key Influence: Upstream CO_2 purity decisively determines downstream energy consumption. ● Core Requirement: Cross-process quality adjustment and full-chain collaborative optimization.
	Jensen et al. [138]	Cost impact analysis of CO_2 impurity tempering	

Aspen HYSYS model to couple the liquid ammonia vaporization cooling capacity with CO_2·BOG heat loads, significantly increasing the re-liquefaction ratio while reducing compression work. Compared to externally supplied cryogenic solutions, this method leverages fuel-side cooling energy, reduces the need for cryogenic equipment and working fluids, enhances heat transfer efficiency, and simplified the process flow that facilitates system integration and control. Fig. 16 shows the variation in liquid ammonia refrigeration energy utilization efficiency under shipboard conditions. Despite load fluctuations, the system maintains stable reliquefaction performance and low energy consumption, balancing both energy efficiency and economic viability.

For the ammonia-based refrigeration pathway, Lee et al. [142] conducted an integrated thermoconformed design evaluation of an ammonia refrigeration-reliquefaction system on a large ocean-going container ship, providing the power requirements and specific energy consumption ranges for the system under various operating conditions. Ji et al. [143] demonstrated through thermodynamic analysis of an NH_3-based vapor compression refrigeration system that this approach significantly improves thermal efficiency and reduces compression power consumption compared to conventional refrigerant schemes during CO_2·BOG reliquefaction. However, both studies share a common limitation: performance validation is based on simulations rather than actual shipboard data, and fuel-side cooling capacity varies with engine load profiles.

MRC delivers stable cooling capacity under cryogenic conditions by optimizing refrigerant mixing ratios. Studies show that for CO_2 with high nitrogen content, the approach proposed by Jeong et al. [144], which combines primary separation with a mixed refrigeration cycle, reduces specific power consumption compared to traditional single-stage ammonia cycles, achieving energy savings of up to 75%. Yan et al. [145] demonstrated through comparative studies that this scheme outperforms single-stage systems in both thermodynamic performance and economic efficiency. Additionally, Lee et al. [146] proposed a novel sub-critical reliquefaction process. Comparative studies with conventional reliquefaction systems demonstrated that, within the practically feasible pressure range, this approach achieves lower specific energy consumption than traditional processes while simplifying the workflow. This improvement enhances system energy efficiency and facilitates engineering integration. As shown in Fig. 17, the specific energy consumption of various reliquefaction systems decreases with increasing LCO_2 storage pressure, indicating that higher LCO_2 storage pressure improves the performance of reliquefaction systems. These studies collectively indicate that MRC efficiency gains are maximized under high-nitrogen conditions. However, refrigerant mixture stability and the complexity of composition control increase operational demands compared to pure-component systems.

Propulsion-reliquefaction integration represents the highest complexity tier, coupling vessel power generation with BOG recovery. Yin and Ju [147] hybridized absorption refrigeration with nitrogen expansion, reducing compression power. Chen et al. [148] integrated ammonia-fueled gas turbines with transcortical CO_2 cycles utilize liquid ammonia cold energy for condensation. This approach eliminates the need for standalone cryogenic equipment. Shen et al. [149] thermodynamically modeled this integration, identifying gas turbine inlet parameter optimization as critical. Cai et al. [150] proposed a liquid hydrogen injector for BOG recovery and reliquefaction. By optimizing the system structure, they enhanced cold energy recovery efficiency and significantly reduced energy consumption. These integrated systems achieve theoretical efficiency maxima but face significant implementation barriers. Propulsion load fluctuations destabilize reliquefaction capacity, while multi-fluid management (NH_3/H_3/CO_2) increases safety and control complexity.

Synthesizing these developments, BOG reliquefaction follows a complexity-efficiency gradient. Fuel-side cooling offers moderate gains with manageable integration, while MRC is suited for high-nitrogen streams but involves composition control overhead. Propulsion

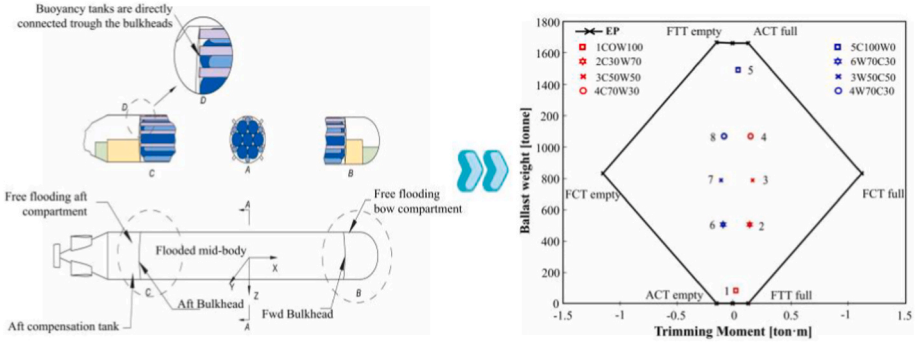


Fig. 15. Subsea Transfer Tanker system designed for liquid CO_2 transport [130].

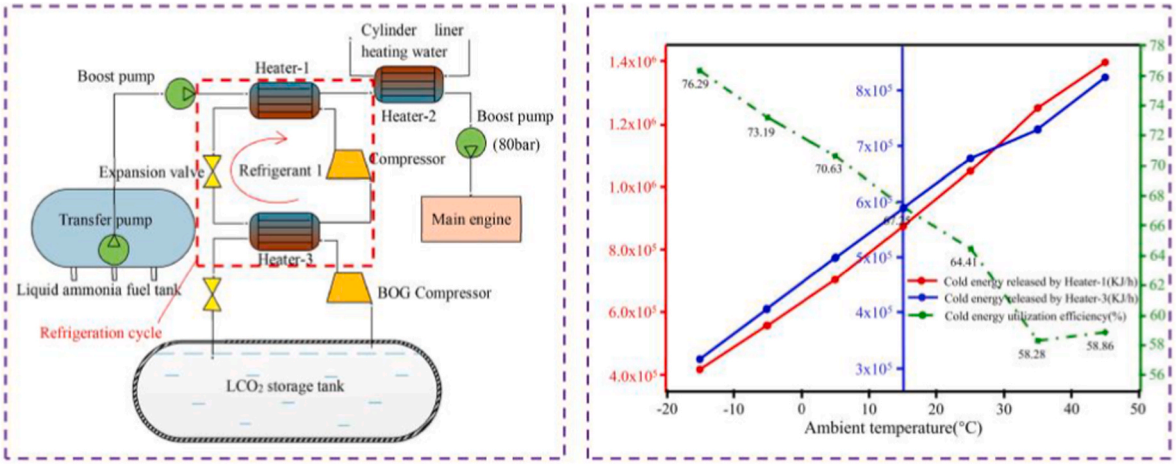


Fig. 16. Variation in liquid ammonia utilization efficiency under shipboard conditions [140].

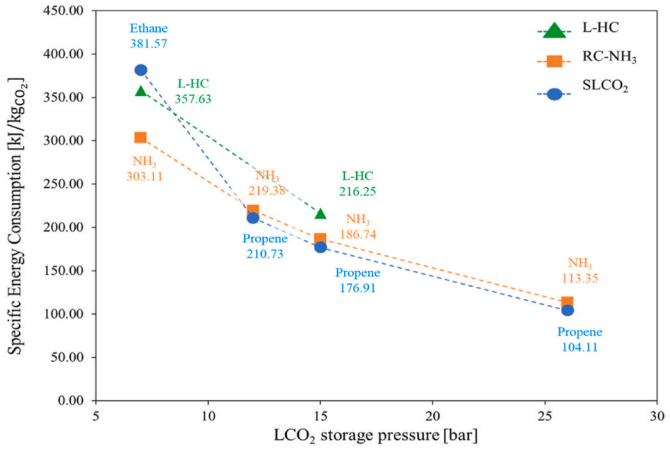


Fig. 17. Effect of LCO_2 storage pressure on unit energy consumption of various reliquefaction systems [142].

integration achieves theoretical optima but faces operational stability challenges.

However, two critical gaps persist. First, there is insufficient cross-system techno-economic assessment. Current studies optimize isolated reliquefaction cycles without quantifying the trade-offs between refrigeration energy savings, propulsion efficiency penalties, and safety system costs. Specific research directions include: (1) integrated vessel-level energy models that couple propulsion, reliquefaction, and auxiliary loads to identify optimal operating modes across voyage profiles; (2) standardized special energy consumption metrics and energy loss decomposition protocols to enable comparative evaluation of fuel-side, MRC, and integrated systems under unified boundary conditions. Second, large-scale demonstration data are lacking. Long-term sea trials under variable ambient temperatures and load profiles are required to validate model predictions and calibrate control strategies for real-world maritime environments.

4.4. Intelligent control and system-level operational optimization scheduling

As the CO_2 liquefaction process becomes more complex with increasing operational steps, traditional control methods that rely on fixed parameters and empirical rules struggle to manage load fluctuations and ship-shore coordination challenges. Recent research focuses on two complementary intelligent strategies: multi-objective optimization using evolutionary algorithms for system design, and AI-driven predictive control for dynamic operation.

Multi-objective optimization at the design level prioritizes energy-cost-efficiency trade-offs through evolutionary algorithms. NSGA-II emerges as the dominant methodological choice across diverse system configurations. Sadni et al. [151] integrated NSGA-II with TOPSIS for multiple organic Rankine cycle (ORC) configurations, identifying R290 as optimal for energy-thermal efficiency balance. Jeong et al. [152] coupled NSGA-II with Aspen HYSYS to optimize a two-stage separation with MRC process for re-liquefaction of nitrogen-containing BOG. The results demonstrated improvements in energy consumption per unit, emissions, and economic performance. These studies demonstrate NSGA-II's versatility in thermodynamic system optimization. However, they differ in decision-making integration: Sadni uses TOPSIS for final selection, while Jeong relies on direct Pareto analysis. Costa et al. [153]

and Gu et al. [154] applied the NSGA-II multi-objective optimization to an integrated CO_2 capture-liquefaction process, shifting optimization scope from standalone liquefaction to upstream-downstream coupling. Also, Yousef et al. [102] developed a multi-objective optimization framework based on genetic algorithms (GA) to synergistically regulate heat exchange networks and pressure parameters, achieving simultaneous reductions in energy consumption and equipment investment. These studies not only demonstrate the broad application of multi-objective optimization across diverse fields but also underscore the importance of balancing multiple objectives in complex system design.

In addition to system design and parameter optimization, current research increasingly focuses on the monitoring and dynamic response control of reliquefaction systems. Predictive control and monitoring address real-time operational stability, with methodological divergence between physics-based and data-driven approaches. Hwang et al. [155] optimized loading process control for pressure-temperature stability, reducing operational fluctuations. Shirmohammadi et al. [65] integrated compression control with thermal load prediction, achieving BOG recovery under daily cycling. These studies rely on thermodynamic models and empirical tuning. In contrast, Han and Yin [156] developed a Koopman-based economic predictive control scheme using deep neural networks with time-varying parameters, representing a shift from physics-based to data-driven modeling. Yin et al. [157] systematically analyzed sub-critical LNG re-liquefaction robustness, providing methodological foundations transferable to LCO_2 systems despite working fluid differences.

The integration of design optimization and predictive control is underdeveloped. Current literature treats them as separate stages: evolutionary algorithms for static design and AI for dynamic operation, with limited feedback between design choices and control architecture. Future research should focus on: (1) co-optimization of process structure and control system design; (2) validating data-driven models under variable maritime load conditions; and (3) standardizing multi-objective weighting methods across industrial applications.

5. Conclusion and outlook

5.1. Conclusion

This review has examined recent progress in CO_2 liquefaction and transportation technologies in context of marine carbon storage. Various liquefaction pathways, such as LH cycles, ARS cycles, and MRC, have been compared with respect to their thermodynamic characteristics and engineering practicality. These studies show that cycle configuration, refrigerant selection, and system integration all play important roles in improving energy efficiency. In parallel, the coupling of external energy sources, including LNG cold energy, industrial waste heat, and renewable energy, provides a promising direction for lowering the overall energy demand of liquefaction and achieving more effective cascading utilization.

In terms of transportation, different CO_2 transport modes exhibit clear differences in safety, operability, and energy performance. Liquid CO_2 shipping remains the most widely discussed option, and the literature consistently highlights the importance of BOG management and recovery technologies for maintaining transport efficiency. Recent work has also explored how optimization methods and advanced control strategies, particularly those incorporating multi-objective thinking or data-driven prediction, can help systems better handle load fluctuations and improve operational stability. Together, these developments outline a technical landscape that is gradually becoming more integrated and more adaptable.

5.2. Outlook and key challenges

Although the field has made notable progress, several issues still require attention. These include how to further reduce energy consumption without compromising economic viability, how to ensure system robustness under variable marine conditions, and how to assess the full-chain performance from capture to storage in a consistent manner. Looking ahead, three directions appear especially important: (1) deeper integration of multiple energy sources to support low-carbon and efficient operation; (2) more refined and adaptive process control to improve system stability and scheduling; (3) closer coordination across the entire CO_2 management chain to avoid isolated optimization. Advances in these areas are expected to provide a clearer and more practical technical route for future large-scale marine CO_2 sequestration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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